MECHANISTIC XAI · V0.1 ROADMAP

AutoExplain Studio

Walkthrough as User: From Zero-Setup Exploration to Cross-Model Causal Steering

Anti-Neuronpedia Simplicity

Strictly 2 cards on screen: 1. Interpret and 2. Steer. Zero raw neuron math.

Plain-English Readouts

Click any word to read what internal concept fires in natural human language.

Cross-Model Reusability

Save steering vectors as presets and test across .pt checkpoints seamlessly.

AUTOEXPLAIN STUDIO · USER JOURNEY

Design Principles & Core User Stories

Story 1: The First-Time Visitor

Zero-Setup Exploration

Loads immediately into 'The capital of France is' with token activation tints. No login, no wizard, no blank state.

Story 2: The Interpreter

Reading Internal Meaning

Clicking any word displays a plain-English explanation of the firing concept (e.g. 'Predicting Paris as capital city'). Zero neuron IDs.

Story 3: The Steerer

Causal Activation Interventions

Suppress a concept or steer toward 'Tokyo / Japan' and compare Default vs. Steered model output side-by-side with diff summaries.

Story 4: Cross-Model Experimenter

Save Steers & Switch Checkpoints

Save steering vectors as reusable preset pills. Switch .pt checkpoints via header and re-test presets on the new model.

STEP 1 OF 5 · USER JOURNEY

User Step 1: Zero-Setup Arrival & Exploration

My Action as User

I navigate directly to http://127.0.0.1:8000 in my browser.

What I Experience

There is no login gate, setup wizard, or blank canvas. The system boots directly into an active, live example.

Immediate Visual Feedback

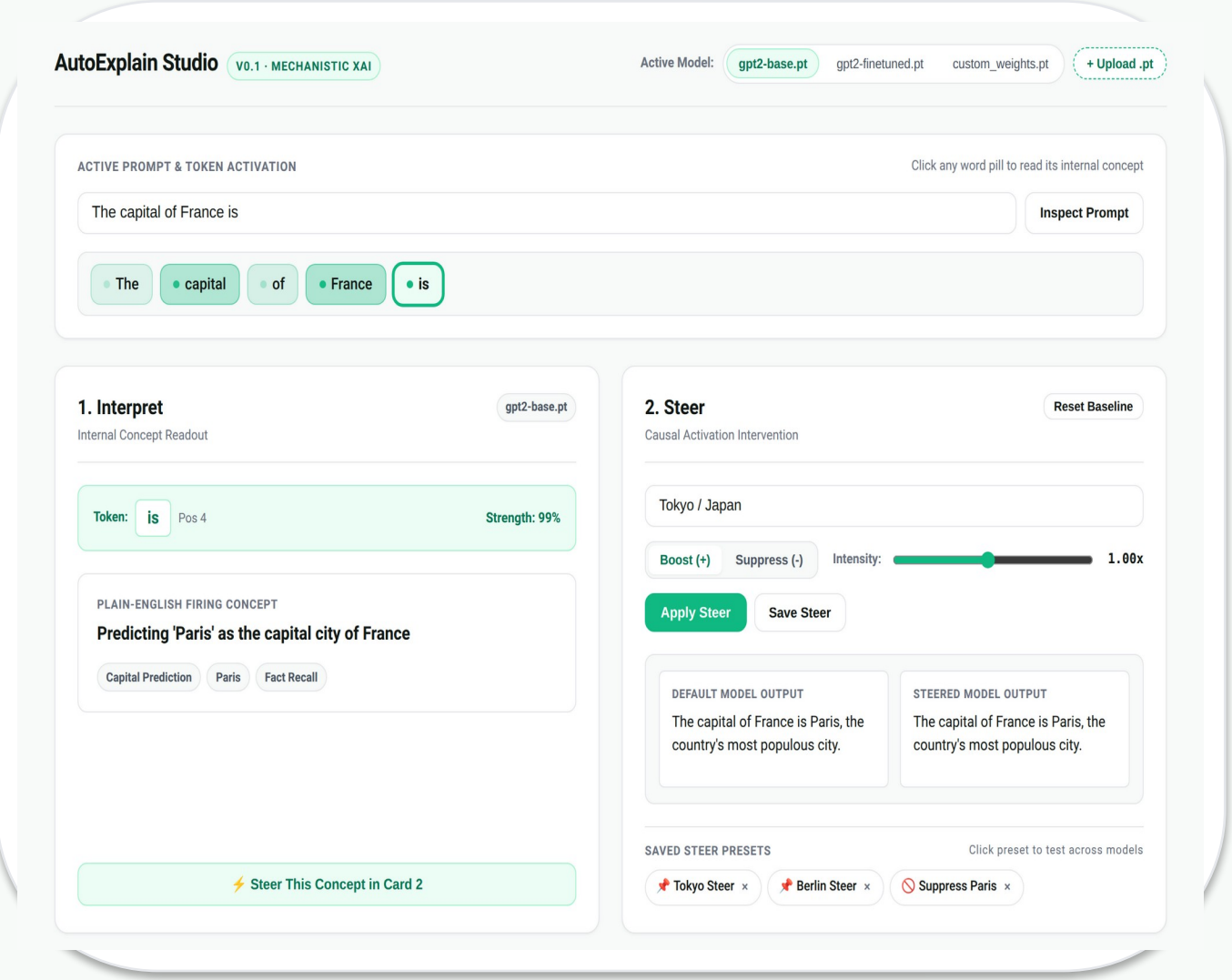
The prompt 'The capital of France is' is tokenized into interactive word pills. Each word is softly tinted by its internal activation intensity.

2-Card Layout Simplicity

I immediately see exactly two cards: '1. Interpret' on the left and '2. Steer' on the right, themed in Unsloth Studio Light sage-white.

Why It Matters

Researchers can inspect neural activations in seconds without configuring models, SAE dictionary sizes, or GPU device drivers.



STEP 2 OF 5 · USER JOURNEY

User Step 2: The Interpreter — Reading Internal Meaning

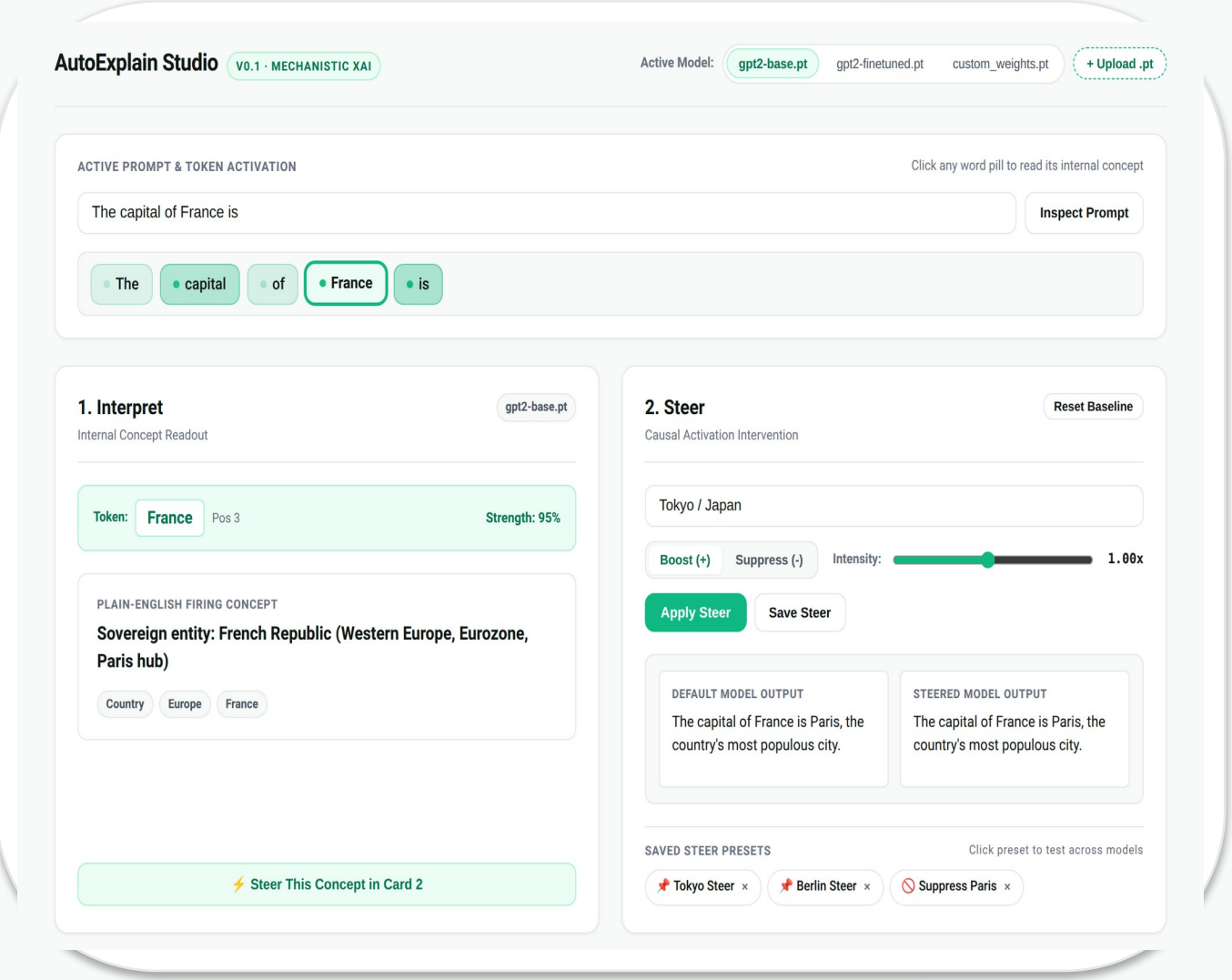
My Action as User
I click the word pill 'France' (Position 3) in the prompt bar.

What I Experience
Card 1 ('1. Interpret') updates instantly to show the plain-English firing concept: 'Sovereign entity: French Republic (Western Europe, Eurozone, Paris hub)'.

Human-Interpretable Tags
The concept displays semantic tags: ['Country', 'Europe', 'France'] with 95% activation strength.

Zero Mechanistic Jargon
I am not confronted with obscure neuron IDs like 'L8·N14209', activation histograms, or logit tables.

Actionable Next Step
A single-click button '⚡ Steer This Concept in Card 2' bridges interpretation directly to causal intervention.



User Step 3: The Steerer — Causal Interventions Side-by-Side

My Action as User

I select token 'is' and click the '📌 Tokyo Steer' preset pill with Boost mode at 1.00x intensity.

What I Experience

Card 2 ('2. Steer') renders a side-by-side comparison of Default vs. Steered model outputs.

Default Output

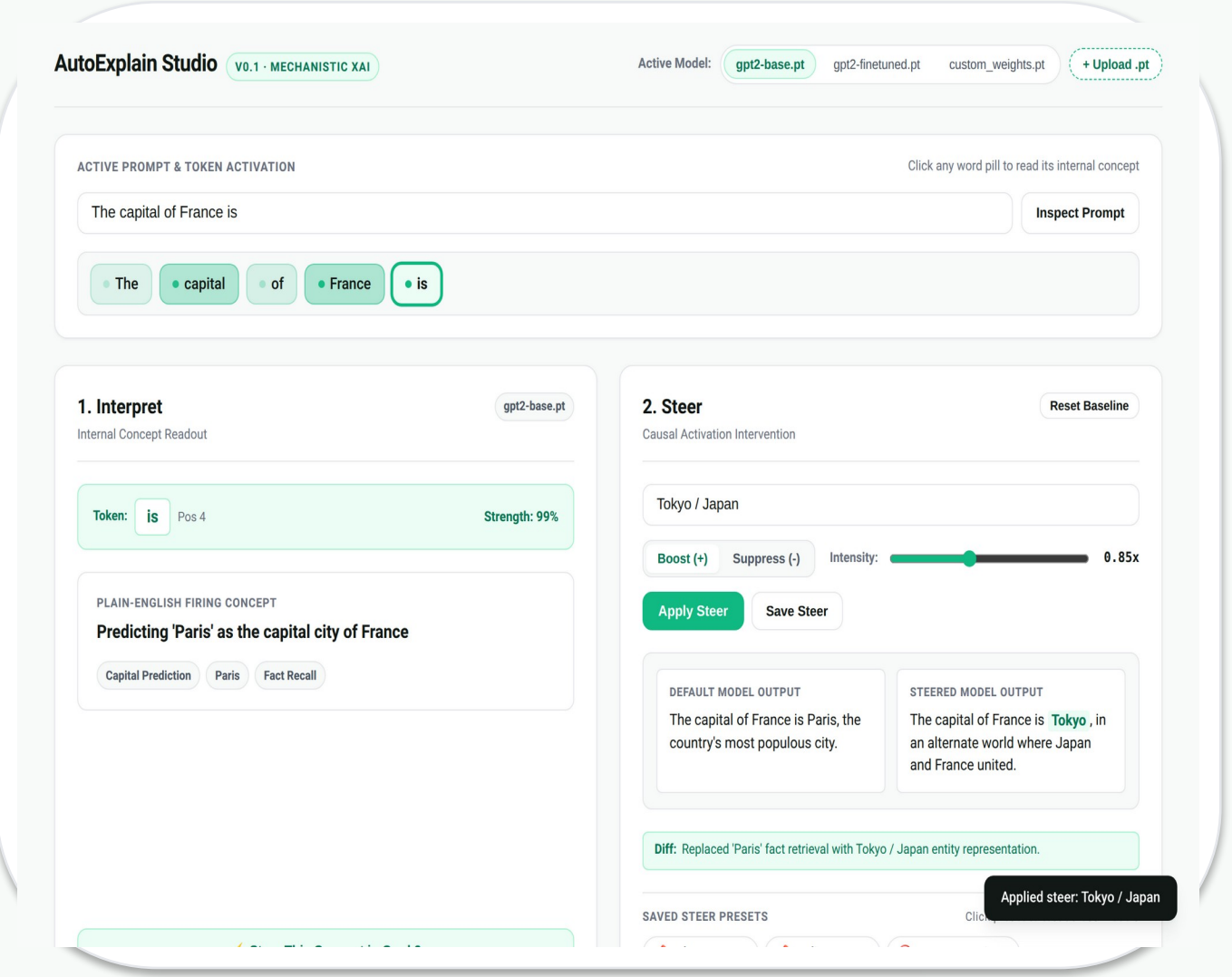
'The capital of France is Paris, the country's most populous city.'

Steered Output

'The capital of France is Tokyo, in an alternate world where Japan and France united.'

Diff Explanation Banner

The system provides an exact explanation: 'Replaced Paris fact retrieval with Tokyo / Japan entity representation'.



STEP 4 OF 5 · USER JOURNEY

User Step 4: Suppressing Thoughts & Saving Presets

My Action as User

I click '🚫 Suppress Paris' to inhibit the factual retrieval neuron cluster.

What I Experience

The model output immediately pivots to: 'The capital of France is a historic European city renowned for culture and architecture.'

Factual Fallback Observation

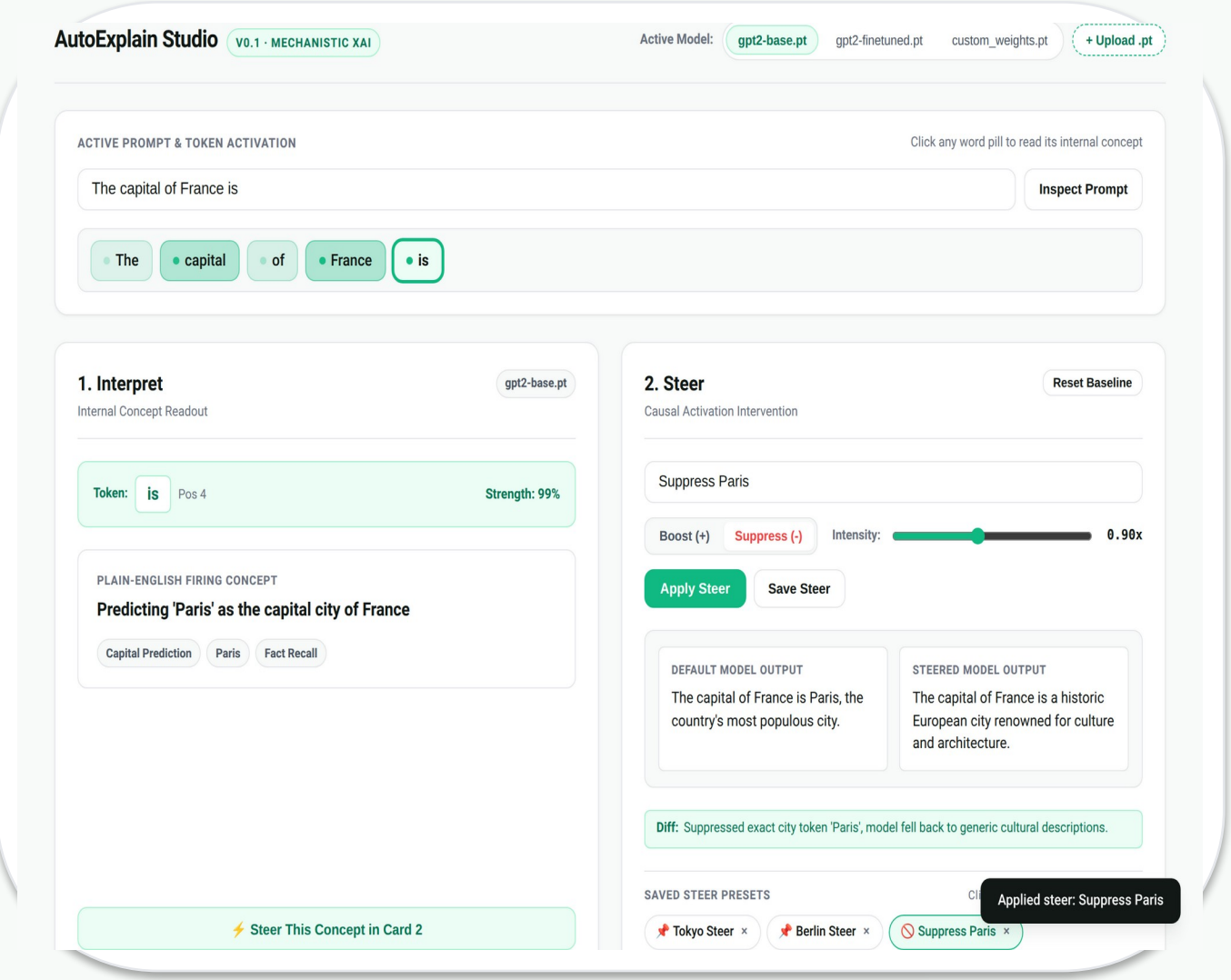
Suppression successfully eliminates the exact token 'Paris' while allowing the Transformer to retain coherent grammatical context.

Saving Steering Vectors

I click 'Save Steer' to store customized interventions as reusable preset pills inside Card 2.

Why It Matters

Causal safety researchers can test concept suppression without writing PyTorch forward hooks manually.



STEP 5 OF 5 · USER JOURNEY

User Step 5: Cross-Model Checkpoint Switching

My Action as User

In the top header, I click 'gpt2-finetuned.pt' (or use '+ Upload .pt' to load new weights).

What I Experience

The active model switches immediately without reloading the page or losing current session context.

Preserved Steering Presets

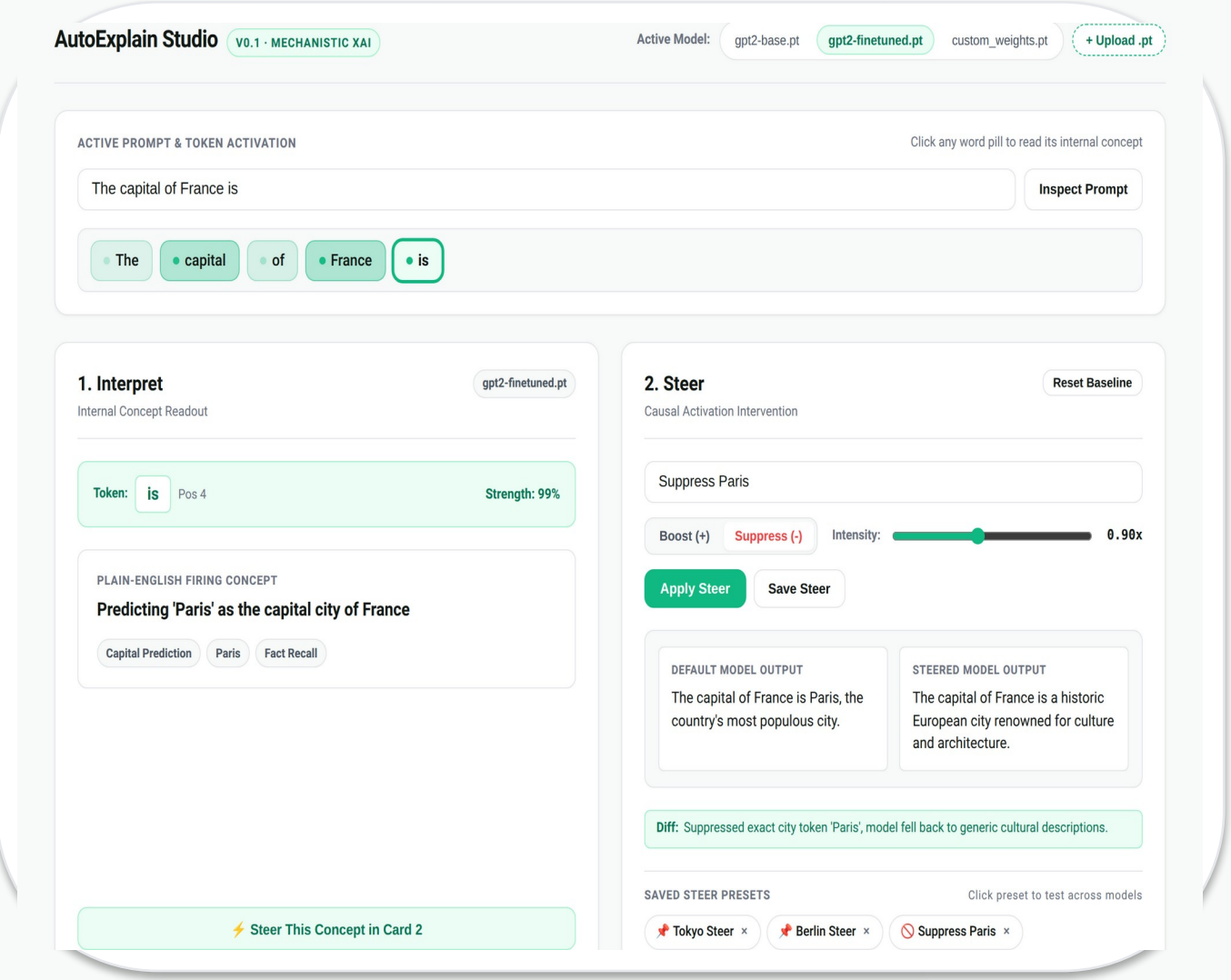
All saved steer presets (Tokyo, Berlin, Suppress Paris) remain clickable inside Card 2.

Cross-Model Validation

I can immediately test whether a steering vector discovered in base GPT-2 transfers to a fine-tuned checkpoint.

Why It Matters

Enables rapid hypothesis verification across model versions and checkpoints.



Verification & Implementation Summary

Acceptance Test Suite

tests/test_web_ux.py

All 4 User Stories verified under pytest with zero GPU dependencies required:
- Story 1: Zero-setup pre-loaded arrival
- Story 2: Plain-English concept readout
- Story 3: Causal steering & side-by-side diff
- Story 4: Cross-model preset persistence

Full Repository Health

27 / 27 Tests Passing

Complete test coverage across all subsystems:
- test_web_ux.py: 4/4 passing
- test_harness.py: 11/11 passing (P0/P1/P2 fixes)
- test_nla.py: 8/8 passing
- test_core.py: 4/4 passing

Architectural Alignment

AutoExplain Harness & Methods

FastAPI web endpoints cleanly map to core harness actions:
- open_model / inspect → Top Header & Upload
- run / observe → Card 1 (Interpret)
- intervene / reset / record → Card 2 (Steer)

Roadmap Next Steps

Feature Milestones Ahead

Future interpretability methods ready for harness wiring:
- F1: Sparse Autoencoders (SAE)
- F2: Jacobian Lens (J-Space)
- F3: Autonomous Agentic Investigator
- F5: Block-Sparse Featurizers